

Lesson Objective

Express the length of an object using centimeter cubes as length units to measure with no gaps or overlaps.

Materials: centimeter cubes, a new crayon, a small glue stick, a dry erase marker

1. Measure the length of an object by lining up centimeter cubes end to end with no gaps or overlaps. How long is the crayon in centimeter cubes? <i>Answer: 9 centimeter cubes long</i>	Place the crayon flat on the table. Tell students that centimeter cubes are all exactly the same size, so we can use them to measure how long the crayon is. Model lining up the first centimeter cube so that one edge matches the endpoint of the crayon. Ask students why it matters that the cube starts at the very end of the crayon. Guide students through laying centimeter cubes along the crayon, end to end, with no gaps and no overlaps, until they reach the other end of the crayon. Have students count the cubes aloud with you. Ask students to state the length of the crayon using the centimeter cube as the length unit.
2. Identify the correct way to measure an object's length using centimeter cubes.	Show students the three pictures one at a time. For each one, ask students what they notice about how the centimeter cubes are placed.

Which picture shows the right way to measure the glue stick? Tell why the other pictures are wrong. Picture A: Centimeter cubes lined up along the glue stick with gaps between some cubes. Picture B: Centimeter cubes lined up along the glue stick, end to end, with no gaps and no overlaps, starting at the endpoint. Picture C: Centimeter cubes lined up along the glue stick with some cubes overlapping each other. <i>Answer: Picture B shows the right way to measure. Picture A has gaps between cubes, so some of the length is not measured. Picture C has cubes that overlap, so some of the length is measured twice.</i>	Ask students which picture shows the right way to measure and have them point to the gaps in Picture A and the overlaps in Picture C. Ask students what happens to the measurement when there are gaps between cubes. Then ask what happens when cubes overlap. Guide students to explain that gaps leave space that does not get measured, and overlaps measure the same space twice. Point out that to measure correctly, every cube must touch the next cube with no gaps and no overlaps, just like in Problem 1.
3. Measure and express the length of an object in centimeter cubes. How long is the dry erase marker in centimeter cubes? <i>Answer: about 12 centimeter cubes long</i>	Place the dry erase marker in front of the student and give them a handful of centimeter cubes. Students work independently to line up centimeter cubes along the marker and state the length.

Monitor For	Instructional Moves
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• Do students line up the first centimeter cube with the endpoint of the object, or do they start partway in? • Do students place each cube touching the next one with no gaps and no overlaps in Problems 1 and 3? • Can students explain that gaps leave some length unmeasured and overlaps measure the same length twice in Problem 2? • Do students count each cube one time and use the last number counted to state the length of the object?	• Before laying out cubes in Problem 1, point to the very end of the crayon and say, "The first cube has to start right here." Have the student match that edge themselves. • If a student leaves gaps between cubes, slide the cubes together and ask, "Is any part of the crayon not under a cube?" so the student names the missing length. • For Problem 2, have the student touch each gap or overlap and count the cubes in each picture aloud to feel the difference. • If a student says a length like "about 12 cubes," remind them we use the word "about" when the object is not exactly a whole number of cubes long.
Reflection Questions • Why do we line up the first centimeter cube with the end of the object? <i>Answer: So we measure the whole length from one endpoint to the other, not just part of it.</i> • What happens to our measurement if there are gaps between the centimeter cubes? <i>Answer: The measurement is too small because the gaps are length that does not get measured.</i> • What happens to our measurement if the centimeter cubes overlap? <i>Answer: The measurement is too big because the same length gets counted twice.</i>	

- How did you know how long the dry erase marker was? *Answer: I counted the centimeter cubes I lined up next to it from one end to the other.*